SOCIOL 723: Social Statistics II

Week 11, Synthetic Control and Its Extensions

Wenhao Jiang
Department of Sociology, Fall 2026

Duke
UNIVERSITY

Today

The Synthetic Control Estimator

Application: Prop 47 and Crime

Extensions

★ = essential, you must understand this ★ = for understanding, not examined

When There Is Only One Treated Unit ★

Difference-in-differences needs a control *group*. What if the treated unit is California, or East Germany, or a single firm?

- Comparative case studies have always done this informally: "compare California to the rest of the country." The choice of comparison was made by the author, and the reader had to trust it.
- Synthetic control makes that choice a data-driven, transparent, pre-registered procedure: construct a weighted average of untreated units that reproduces the treated unit's pre-treatment trajectory.
- "Arguably the most important innovation in the policy evaluation literature in the last 15 years" (Athey and Imbens 2017).

The identifying idea is not new. What is new is that the weights are chosen by an algorithm, reported in a table, and checked against placebo units.

Synthetic Control

Setup ★

Unit 1 (California) is treated from year $T_0 + 1$ onward. Units $j = 2, \ldots, J + 1$, the donor pool, are never treated. For each year $t > T_0$ the effect is

$$\tau_t = Y_{1t}(1) - Y_{1t}(0).$$

We observe $Y_{1t}(1)$; the whole problem is the number $Y_{1t}(0)$: California's cigarette sales in, say, 1995 had Prop 99 never passed.

Every method today has the same form

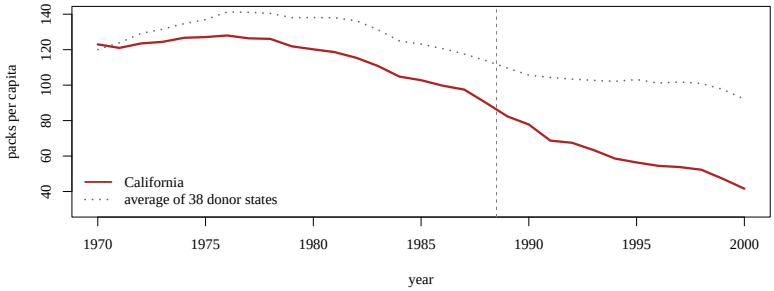
Predict $Y_{1t}(0)$ by a weighted sum of the donors' outcomes in the same year,

$$\hat{Y}_{1t}(0) = w_2 Y_{2t} + w_3 Y_{3t} + \cdots + w_{J+1} Y_{J+1,t} = \textstyle\sum_{j=2}^{J+1} w_j Y_{jt},$$

and estimate $\hat{\tau}_t = Y_{1t} - \hat{Y}_{1t}(0)$. Methods differ only in how they choose the weights w_j.

Running Example: Proposition 99

California raised its cigarette tax by 25 cents a pack in 1989 and funded anti-smoking campaigns. Outcome: packs sold per capita. Donors: 38 states with no large tobacco program or tax increase in the period.



- California falls faster than the donor average, and it was already doing so before 1989 (1988: 90.1 against 113.8) (figs/w11_scalar.R).
- How much of the post-1989 gap is the law? Start with the naive answer.

The Naive Approach: the Donor Average ★

Give every donor the same weight, $w_j = 1/38$, and add California's average pre-1989 gap: this is DiD (Week 10) (figs/w11_scalar.R).

	1970–88 mean	1989–2000 mean	change
California	116.2	60.4	−55.8
average of 38 donors	130.6	102.1	−28.5
difference	−14.4	−41.7	**−27.3**

- DiD assumes California would have fallen as fast as the donor average.
- It was not doing so before the law: from 1970 to 1988 California fell by 32.9 packs, the donor average by 6.3. Parallel trends fails in the pre-period, so −27.3 is too large in absolute value.
- The fix: choose the weights so that the donors' weighted past *looks like* California's past.

Two Donors, One Weight ★

Start with two donors, Utah and Nevada. One number to choose, w:

$$\hat{Y}_{\mathrm{CA},t}(0) = w\,Y_{\mathrm{UT},t} + (1-w)\,Y_{\mathrm{NV},t}, \qquad 0 \le w \le 1.$$

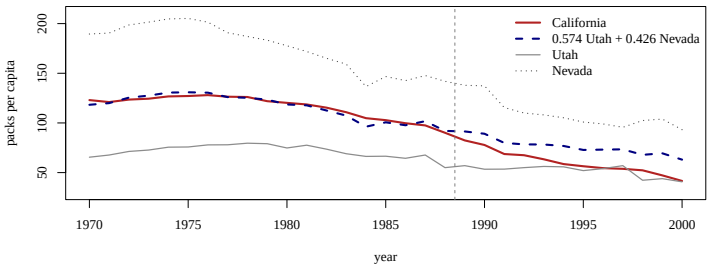
Pick w to make the pre-period gaps as small as possible:

$$\begin{aligned}
\mathrm{SSR}(w) &= \textstyle\sum_{t=1970}^{1988} \left(Y_{\mathrm{CA},t} - w\,Y_{\mathrm{UT},t} - (1-w)\,Y_{\mathrm{NV},t}\right)^2 && \text{squared pre-period gaps} \\
&= \textstyle\sum_t \left[(Y_{\mathrm{CA},t} - Y_{\mathrm{NV},t}) - w\,(Y_{\mathrm{UT},t} - Y_{\mathrm{NV},t})\right]^2 && \text{regroup the } w \text{ terms} \\
0 &= -2\textstyle\sum_t (Y_{\mathrm{UT},t} - Y_{\mathrm{NV},t})\left[(Y_{\mathrm{CA},t} - Y_{\mathrm{NV},t}) - w\,(Y_{\mathrm{UT},t} - Y_{\mathrm{NV},t})\right] && \text{derivative set to 0} \\
w &= \frac{\sum_t (Y_{\mathrm{UT},t} - Y_{\mathrm{NV},t})(Y_{\mathrm{CA},t} - Y_{\mathrm{NV},t})}{\sum_t (Y_{\mathrm{UT},t} - Y_{\mathrm{NV},t})^2} && \text{solve for } w
\end{aligned}$$

- This is OLS through the origin: regress $Y_{\mathrm{CA},t} - Y_{\mathrm{NV},t}$ on $Y_{\mathrm{UT},t} - Y_{\mathrm{NV},t}$, one observation per pre-period year.
- SSR is a parabola in w: if the solution falls outside $[0, 1]$, take the nearer endpoint.

Two Donors on Prop 99

$w = 123{,}200/214{,}460 = 0.574$ (figs/w11_scalar.R).



- SSR: $w = 0$ (Nevada only) 70,976; $w = 1$ (Utah only) 39,037; $w = 0.5$ 1,392; $w = 0.574$ 202. Neither state alone resembles California; the mixture does.
- Pre-period fit in packs: root mean squared prediction error, $\text{RMSPE} = \sqrt{\text{SSR}/19} = \sqrt{202/19} = 3.26$, the typical size of a pre-period gap.
- Effect: average gap 1989–2000 $= -15.8$ packs, against -27.3 from the donor average.

Many Donors ★

Same objective, one weight per donor:

$$\min_{w_2,\ldots,w_{J+1}} \sum_{t\le T_0} \Big(Y_{1t} - \sum_{j=2}^{J+1} w_j Y_{jt}\Big)^2 \qquad \text{subject to} \quad w_j \ge 0, \quad \sum_{j=2}^{J+1} w_j = 1.$$

- The same two rules as with Utah and Nevada, now on every weight: no weight is negative, and the weights add to one. With two donors the weights w and $1-w$ already add to one, so the rules left only $0 \le w \le 1$.
- With 38 donors there is no formula like the one for w; software finds the weights (a quadratic program).
- On Prop 99, all 19 pre-years, 38 donors: average gap 1989–2000 $= -19.5$, pre-period RMSPE 1.66 (figs/w11_prop99.R).
- Without the constraints, 38 weights could fit 19 years exactly; more on this below.

Why the Constraints Matter ★

The restrictions $w_j \geq 0$ and $\sum_j w_j = 1$ look like technicalities. They are the heart of the method.

- **No extrapolation**: the synthetic unit is a *convex combination* of real units, so it lies inside the donor pool's support. Unconstrained regression would happily assign weights of $+3$ and -2 and predict outside anything ever observed.
- **Sparsity**: the constraints usually produce weights on a handful of donors and zero on the rest. You can print the weight table and let readers judge whether the comparison is sensible.
- **Transparency**: the counterfactual is a named, weighted set of real places, not a regression coefficient.
- If no convex combination matches the treated unit, if California is richer than every donor, the method tells you, through poor pre-treatment fit. That is a feature.

What the Constraints Prevent, on Prop 99

Drop $w_j \geq 0$ and keep $\sum_j w_j = 1$: with 38 donors and 19 pre-years, infinitely many weightings fit the pre-period exactly. Take the smallest one (minimum norm). Both columns fit the 19 pre-period outcomes (figs/w11_prop99.R):

	SC weights	regression weights
nonzero donors	6	38
negative weights	0	17 (as low as −0.13, Tennessee)
pre-period fit	RMSPE 1.66	exact
average gap 1989–2000	−19.5	−15.5

- An exact fit with $J > T_0$ is guaranteed and therefore uninformative: it fits the noise.
- Negative weights mean the counterfactual California is "West Virginia plus Connecticut minus Tennessee minus Mississippi": extrapolation no reader can inspect (Abadie 2021).

Matching Predictors as Well as Past Outcomes ★

Abadie, Diamond, and Hainmueller (2010) match *predictors* $k = 1, \ldots, K$. Each X_{jk} is *one number* per state that the researcher defines: mean price 1980–88, sales in 1975, ...

Inner step. Fix predictor weights $v_k \geq 0$. Choose w to match the predictors:

$$w(v) = \underset{w_2,\ldots,w_{J+1}}{\arg\min} \ \sum_{k=1}^{K} v_k \Big(X_{1k} - \sum_{j=2}^{J+1} w_j X_{jk}\Big)^2, \qquad w_j \geq 0, \ \ \sum_j w_j = 1.$$

Outer step. Among all choices of v, keep the one whose $w(v)$ fits the pre-period *outcomes* best:

$$\min_{v} \ \sum_{t \leq T_0} \Big(Y_{1t} - \sum_j w_j(v)\, Y_{jt}\Big)^2.$$

- w minimizes the *predictor* gaps; the outcome fit only chooses v.
- Two kinds of years: the outer step uses *every* pre-year, 1970–88; predictors summarize windows the researcher picks (a choice).

The Two Steps on Two Donors ★

Utah and Nevada again; two covariates (figs/w11_scalar.R): price, 1980–88 (CA 89.42, UT 89.43, NV 93.24); beer sales, 1984–88 (CA 24.28, UT 13.34, NV 37.00).

v_{price} ($v_{\text{beer}} = 1 - v_{\text{price}}$)	inner $w(v)$	outcome SSR
0 (beer only)	0.538	494
0.5	0.549	337
0.768	**0.574**	**202**
1 (price only)	1.000	39,037

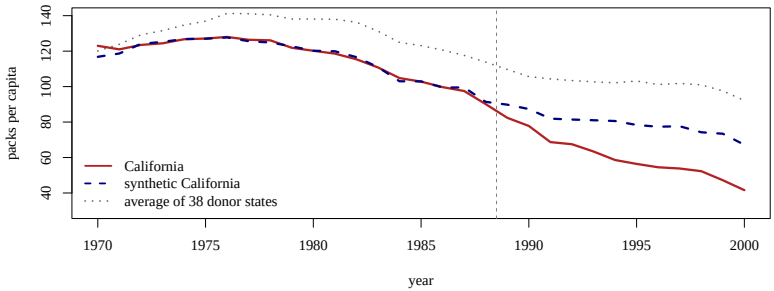
w = Utah's weight; Nevada gets $1 - w$.

- The loop: for $v_{\text{price}} = 0, .001, \ldots, 1$, get $w(v)$, then the outcome SSR; keep the smallest of the 1,001 tries.
- Price only: Utah has California's price, so $w = 1$. Beer only: $w = 0.538$.
- v weights covariates, w weights donors; the outcome only chooses v.

v depends on units (price varies little, so it needs a big v): a tuning knob, not importance.

Synthetic California

ADH's seven predictors (four covariates; packs sold in 1975, 1980, 1988), v from the outer step: Utah .34, Nevada .24, Montana .21, Colorado .15, Connecticut .06, applied to every year (figs/w11_prop99.R).



- Tracks California to 1988 (RMSPE 1.78), then separates. Average gap 1989–2000: −18.8 packs (about −24%); in 2000: −25.6.
- Outcomes-only weights (Many Donors slide) give −19.5; DiD gave −27.3.

Searching for v When There Are Many Predictors

With two predictors, v had one free number and we tried 1,001 values. With ADH's seven, v has six free numbers (they add to 1).

- A grid of 1,000 values each would be 10^{18} tries. Software searches: move v in the direction that lowers SSR until it stops falling.
- The search can stop at a *local* minimum; different software reports different v and w for the same data (Klößner et al. 2018). Report w and the fit; do not read v_k as importance.
- A common way out: drop the covariates and fit w to the pre-period outcomes, as on the Many Donors slide. No v, nothing to search. Prop 99: SSR 52 (ADH's 60), gap -19.5.
- Covariates guard against fitting noise in the outcome path; past outcomes guard against unmeasured traits. Report both when they differ.

What Have We Assumed? ★

Everything so far used 1970–88 only. After 1988 we keep the same weights:

$$\hat{Y}_{\text{CA},1995}(0) = .342\,Y_{\text{UT},1995} + .238\,Y_{\text{NV},1995} + .209\,Y_{\text{MT},1995} + .149\,Y_{\text{CO},1995} + .062\,Y_{\text{CT},1995}.$$

Assumption: a mix that tracked California for 19 years would have kept tracking it.

Why would a mix track California before 1989? Two possible reasons.

1. **It shares what drives California's sales**: income, prices, culture, how health-conscious people are. Then it keeps tracking after 1988. This is what we hope.
2. **Luck**: with 38 donors, some mix can cancel California's year-to-year bumps. Those bumps are new after 1988, so the tracking stops.

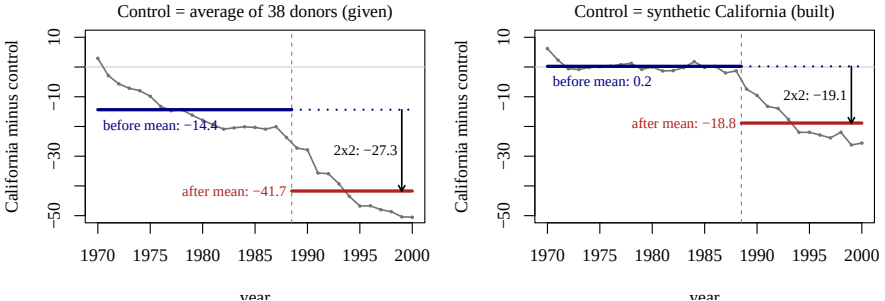
Synthetic Control Is a 2×2 With a Built Control ★

DiD and SC both compare California's change with a control's change, before (1970–88) and after (1989–2000). Mean packs per capita; differences use unrounded means (figs/w11_2x2.R).

	before	after	after − before
California	116.2	60.4	
Control 1: average of 38 donors (*given*)	130.6	102.1	
gap, California − control 1	−14.4	−41.7	**−27.3** (DiD)
Control 2: synthetic California (*built*)	116.0	79.2	
gap, California − control 2	0.2	−18.8	**−19.1** (SC)

- Same 2×2, different control: DiD takes it as given; SC builds it so that the before gap is about 0.
- Parallel trends, for both: without the law, the gap stays at its before level.
- SC's usual estimate, the after gap alone (−18.8), takes the before gap as 0.

The Two 2×2s on Prop 99 ★



- Gray: the yearly gap. Blue: before mean; dotted: where parallel trends says it stays without the law. Red: after mean. 2×2 = red − blue.
- Left: the gap fell before the law (+2.9 in 1970, −23.7 in 1988), so parallel trends already fails; −27.3 includes that trend.
- Right: the built control's gap is near 0 before 1989; the assumption is that it stays near 0.

When Parallel Trends Fails After Weighting ★

Weighting does not remove the assumption: without the law, California and synthetic California would have moved together after 1988. It fails when:

1. **The fit was luck**: with many donors and a short pre-period, the weights match noise, not what drives sales.
2. **No mix reaches California**: it is extreme; the fit fails before 1989.
3. **A donor has its own shock after 1988**: e.g. its own cigarette tax.
4. **The law reaches the donors**: Californians buy cigarettes in Nevada (weight .24), so Nevada's sales rise because of Prop 99. Synthetic California is then too high: the effect is overstated.
5. **California reacts early**: once the law is known, before it starts.
6. **Something else changes in California at the same time**: SC estimates all of it. Prop 99 was itself a tax rise *and* an anti-smoking program; SC cannot split them.

What to Check, What to Do ★

- **Choose the donors first**: drop states with their own tobacco policies (ADH dropped 11 states and DC) and neighbours the law may reach. Then accept the fit; do not shop for the pool that fits best.
- **Pre-period fit**: plot both paths for 1970–88; report the RMSPE in packs.
- **In-time placebo**: pretend the law began in 1981, fit the weights on 1970–80, and check that the 1981–88 gap is near 0.
- **Leave-one-out**: drop each donor with positive weight, refit; the estimate should barely move.
- **Anticipation**: a law can matter once it is known. Prop 99 passed in November 1988 and the tax rose in January 1989, so smokers could stock up first. If the gap opens early, start the treatment at the vote.

Inference: the Placebo Test ★

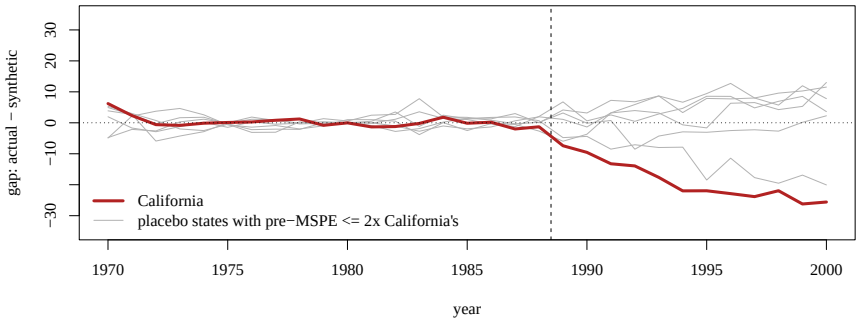
One treated state, so no usual standard error. Instead, run the method on every state (Abadie, Diamond, and Hainmueller 2010):

1. Treat each of the 39 states in turn as "treated"; build its synthetic control from the other 38.
2. Compute its ratio $\text{RMSPE}^{\text{after}}/\text{RMSPE}^{\text{before}}$. Dividing by the pre-period fit keeps a state the method could not fit from looking like a big effect.
3. Rank California among the 39: $p = \text{rank}/39$.

Can it be reported as a p-value? Yes: it is a permutation test, and papers report it as the p-value.

- Meaning: if the law did nothing and California were one of 39 interchangeable states, a ratio this large would come up 1 time in 39. Exact only if the treated state had been picked at random.
- Always state the floor: with 39 states the smallest possible p is $1/39 = 0.026$.

Placebos on Prop 99



- Pre-period MSPE: California 3.17; median donor 15.4. Only 6 donors fit as well as twice California's MSPE (shown).
- Post/pre MSPE ratio: California 124, largest of 39 (next: Georgia, 47). Placebo rank $1/39 = 0.026$, the smallest attainable.

A Standard Error From the Placebos

Second option (Arkhangelsky et al. 2021, `synthdid`): estimate the "effect" for placebo states, and use their spread as the standard error.

$$\text{SE} = \text{sd of the placebo estimates}, \qquad 95\%\ \text{CI} = \hat{\tau} \pm 1.96\,\text{SE}.$$

Prop 99	estimate	placebo SE	95% CI
SC	−19.6	11.1	[−41.4, 2.1]
SDiD	−15.6	8.9	[−33.1, 1.9]

(figs/w11_prop99.R)

- The SC interval includes 0, while the rank test gives $p = 0.026$. The SE uses the raw placebo effects, so states the method fits badly inflate it; the ratio discounts them.
- Report the rank p-value; add the placebo SE if you want an interval, and say which one you used.

In-Time Placebo and Leave-One-Out on Prop 99

In-time placebo: pretend the law began in 1981; fit on 1970–80 with ADH's predictors moved back (covariates 1972–80; sales in 1970, 1975, 1980) (figs/w11_prop99.R).

- Fit 1970–80: RMSPE 0.89. Average gap 1981–88: −2.9 packs.
- Not zero: California was already pulling away in the 1980s, by about 15% of the post-1989 effect (−18.8). Report it.

Leave-one-out: drop each donor with positive weight, refit.

dropped	Utah	Nevada	Montana	Colorado	Connecticut
average gap	−18.2	−20.6	−17.6	−19.3	−18.7

No single donor drives the estimate.

In the Literature: German Reunification

Abadie, Diamond, and Hainmueller (2015)

The question: what did reunification in 1990 cost West Germany? GDP per capita (PPP, 2002 USD); 16 OECD donors; predictors trained on 1971–80, validated on 1981–90.

donor	weight
Austria	0.42
United States	0.22
Japan	0.16
Switzerland	0.11
Netherlands	0.09

- About −1,600 USD per person per year, 1990–2003; 2003 GDP per capita 12% below synthetic.
- In-time placebo, reunification in 1975: no gap.
- Post/pre RMSPE ratio about 16, largest of 17: placebo rank $1/17 \approx 0.06$.
- Leave-one-out: dropping the US gives the smallest effect, about −630 USD per year.

Replications across software packages and variable orderings do not reproduce the published weights (Klößner et al. 2018): the predictor weights v_k are poorly identified.

Prop 47

In the Literature: Bartos and Kubrin (2018)

The question. California's Proposition 47 (November 2014) reclassified some drug and property felonies as misdemeanors, shrinking prison and jail populations. Did crime rise? Opponents pointed to a 2015 uptick.

The data. FBI Uniform Crime Reports, state rates per capita, 1970–2015: seven offenses, each with its own synthetic California. 44 pre-period years; one post-period year (2015).

The design. Donor pool: the other 49 states, since none adopted a comparable reform. Weights from pre-period crime trends. Effect: the 2015 gap.

Decision rule. A gap smaller than the pre-period RMSPE is treated as no effect; a placebo rank in the top 5 of the 50 states (share 0.10) counts as evidence.

Bartos and Kubrin: Results

offense	2015 gap	placebo rank
homicide, rape, aggravated assault, robbery	within pre-period RMSPE	—
burglary	within pre-period RMSPE	—
larceny	under 10% above synthetic	4 of 50 (0.08)
motor vehicle theft	about 20% above synthetic	13 of 50 (0.26)

- Violent crime: no detectable effect. Larceny: a modest increase, weakly distinguished from placebo variation.
- Leave-out check: the larceny gap disappears when the heaviest donors are removed.

Extensions

Penalized Synthetic Control (Abadie and L'Hour 2021)

Penalize the distance to every donor used (sums over predictors $k = 1, \ldots, K$ and donors $j = 2, \ldots, J+1$; $w_j \geq 0$, $\sum_j w_j = 1$):

$$w^{\text{pen}} = \underset{w_2,\ldots,w_{J+1}}{\arg\min} \underbrace{\textstyle\sum_k \left(X_{1k} - \sum_j w_j X_{jk}\right)^2}_{\text{the mix vs. California}} + \lambda \textstyle\sum_j w_j \underbrace{\sum_k \left(X_{1k} - X_{jk}\right)^2}_{\text{donor } j \text{ vs. California}}.$$

Hypothetical, $K = 1$: $X_1 = 5$; donors A $= 4$, B $= 6$, C $= 0$, D $= 10$ (figs/w11_scalar.R).

mix	$\sum_j w_j X_j$	mix vs. California	penalty $\sum_j w_j (X_1 - X_j)^2$
$\frac{1}{2}$A $+\frac{1}{2}$B	5	0	$\frac{1}{2} \cdot 1 + \frac{1}{2} \cdot 1 = 1$
$\frac{1}{2}$C $+\frac{1}{2}$D	5	0	$\frac{1}{2} \cdot 25 + \frac{1}{2} \cdot 25 = 25$

- Both fit perfectly; plain SC cannot choose. Any $\lambda > 0$ picks the nearby A, B.
- $\lambda \to 0$: SC; $\lambda \to \infty$: nearest-neighbour matching (Week 6).
- `pensynth::cv_pensynth` picks λ by pre-period fit. Prop 99: $\lambda \approx 0$, the same weights as SC (Lab 11).

Augmented Synthetic Control (Ben-Michael, Feller, and Rothstein 2021)

Idea: if synthetic California misses California before 1989, part of the later gap is that mismatch, not the law. Predict it and subtract it. T: one post-treatment year (1989, . . . , 2000); repeat for each and average.

1. Build SC as usual: weights w_j.
2. Pool the 38 donors and regress year-T sales on each state's own 19 pre-period sales, $Y_{jT} \approx \eta_0 + \sum_{t \le 1988} \eta_t Y_{jt}$ (ridge; no unit or year effects).
3. Predict each state from its own past: $\hat{m}_{jT}$ (donors), $\hat{m}_{1T}$ (California).
4. Subtract the gap their pasts alone predict:

$$\text{effect}_T = \underbrace{\left(Y_{1T} - \textstyle\sum_j w_j Y_{jT}\right)}_{\text{SC gap}} - \underbrace{\left(\hat{m}_{1T} - \textstyle\sum_j w_j \hat{m}_{jT}\right)}_{\text{gap the pasts predict}}.$$

- Perfect pre-period fit: same pasts, same predictions, nothing subtracted.
- Prop 99 (`augsynth`): SC gap −19.5, predicted gap −3.6, augmented −16.0 (figs/w11_prop99.R).

Synthetic DiD: Match the Trend, Not the Level

SC needs California's *level* in every pre-year; **DiD** needs only its *trend*. **Synthetic DiD** (Arkhangelsky et al. 2021) builds the control like SC but asks only for the trend:

$$\min_{\omega,\ c} \ \sum_{t \le 1988} \left(Y_{1t} - c - \sum_j \omega_j Y_{jt} \right)^2, \qquad \omega_j \ge 0, \ \sum_j \omega_j = 1.$$

The free constant c lets the control sit above or below California: only the *shape* of the path has to match. What this buys:

- **Extreme units**: if California had the lowest sales, no average of donors could reach its level, but a mix can still share its trend.
- **More mixes qualify**, so the weight can spread over many states (a small penalty encourages this).
- **The cost**: the level gap is assumed constant, so we are back to DiD's parallel trends, now for a weighted control.

Synthetic DiD on Prop 99

Two sets of weights, then the 2×2 (figs/w11_prop99.R):

- **Donor weights** ω_j, fitted with the free constant c: Nevada .12, New Hampshire .11, Connecticut .08, …
- **Pre-year weights** λ_t: the "before" period is the pre-years that best predict the donors' post-period mean: 1986–88 (.37, .21, .43). The level gap is measured on years like the after period.

	before (weighted)	after	after − before
California	95.1	60.4	
SDiD control	120.5	101.3	
gap	−25.4	−41.0	**−15.6**

- The control sits 25 packs above California before 1989. SC would not allow that; SDiD does, because only the trend has to match.
- DiD −27.3, SC −19.6, SDiD −15.6 (`synthdid`, as in their Table 1).

Choosing Among Them

Situation	Method	`R`
One treated unit, good pre-fit	classic synthetic control	`Synth`, `tidysynth`
One treated unit, poor pre-fit	augmented / ridge SC	`augsynth`
Many units, want DID robustness	synthetic DID	`synthdid`
Many units, staggered, covariates	Callaway–Sant'Anna (Wk 10)	`did`

- These are not rivals so much as points on a spectrum between "trust the weights" and "trust the outcome model." The augmented versions deliberately use both, the same insurance logic as AIPW.
- Report the classic estimator alongside whichever you prefer, so readers can see how much the extrapolation is doing.

Looking Ahead

- Synthetic control completes the design-based half of the course. Weeks 5–11 have all been variations on one question: *where does the counterfactual come from?*
- Notice the recurring structure: choose weights on comparison units (Week 6), choose an exogenous slice of variation (Weeks 7, 9), choose a differencing scheme (Week 10), choose a weighted donor pool (Week 11).
- **Weeks 12–13** change the question again: what happens when prediction rather than identification is the goal, and what, if anything, flexible prediction can contribute back to causal work.

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